दक्षिण बिहार केन्द्रीय विश्वविद्यालय

CENTRAL UNIVERSITY OF SOUTH BIHAR

Question Paper of End-Term Examination-May 2025

Name of Department: Computer Science | Programme: M.Sc(AI)

Session: 2024-26 | Semester: II

Course Title: Machine Learning | Course Code: CAI82DC01004

Course Credit: 4 | Duration: 3 Hours

Enrollment Number: | Max Marks: 70

There are three sections (A, B, and C). Answer all questions as per instructions. Points against each question are mentioned in the parenthesis.

Section - A

Answer all the questions in not more than 50 words. Each question carries 1 Points.
($1 \times 10 = 10$ Points)

1. Define the task and performance measure for checkers' learning problem.
2. Write the names of the different components of designing a learning system.
3. Let h_i and h_k be the boolean valued functions defined over X. Write the condition that states that h_i is more general than h_k.
4. Write the expressions for the Entropy of a sample S, and the Gain of an attribute A with respect to a sample S.
5. Define overfitting of a hypothesis.
6. Define $sgn(y)$ function.
7. Let $Y = \psi(X)$ where $Y = (Y_1, \ldots, Y_m)$ and $X = (X_1, \ldots, X_n)$ then define $\frac{\partial Y}{\partial X}$.
8. Write the variance and standard deviation of a binomial random variable X with parameters n and p.
9. State Bayes theorem.
10. In the case of Bayesian learning, define MAP and ML hypothesis.

Section - B

Answer ANY FIVE questions in not more than 300 words. Each question carries 6 Points.
($6 \times 5 = 30$ Points)

1. Explain the Linear regression model of learning.
2. Assume the letters A, B, C and D (output) are the attributes and each of these attributes takes two values. Write the gain expression for each of the input attributes. Their is no need to compute log values, maintain the log values as it is.

A	B	C	D
a_1	b_1	c_1	Y
a_1	b_2	c_1	N
a_1	b_2	c_2	N
a_2	b_2	c_2	Y

Table 1: Table for Section B Question 2

3. For an input $X^T = (X_1, \ldots, X_p)$ we define $f(X) = \beta_0 + \sum_{i=1}^{p} X_i\beta_i$. Write this definition in the form of matrix multiplication. Express $RSS(\beta) = \sum_{i=1}^{N}(y_i - x_i^T\beta)^2$ where y_i is a scalar, $\beta^T = (\beta_0, \cdots \beta_p)$, and $x_i^T = (1, x_{i1}, \cdots x_{ip})$ in the form of matrix multiplication.

4. Compute the differential of the following functions which we generally use in multilayer neural network.

   (a) $\sigma(x) = \frac{1}{1+e^{-x}}$ [3 marks]

   (b) $\sigma(x) = tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$ [3 marks]

5. Describe K-means algorithm of clustering.

6. Explain the backpropogation algorithm in detail for a multilayer neural network that contains an output layer and one hidden layer.

Section - C

Answer ANY TWO questions in not more than 1000 words. Each question carries 15 Points.

(2 × 15 = 30 Points)

1. Explain candidate elimination algorithm.

2. Describe the decision tree learning algorithm with an example.

3. Describe the idea of Bayesian belief network.